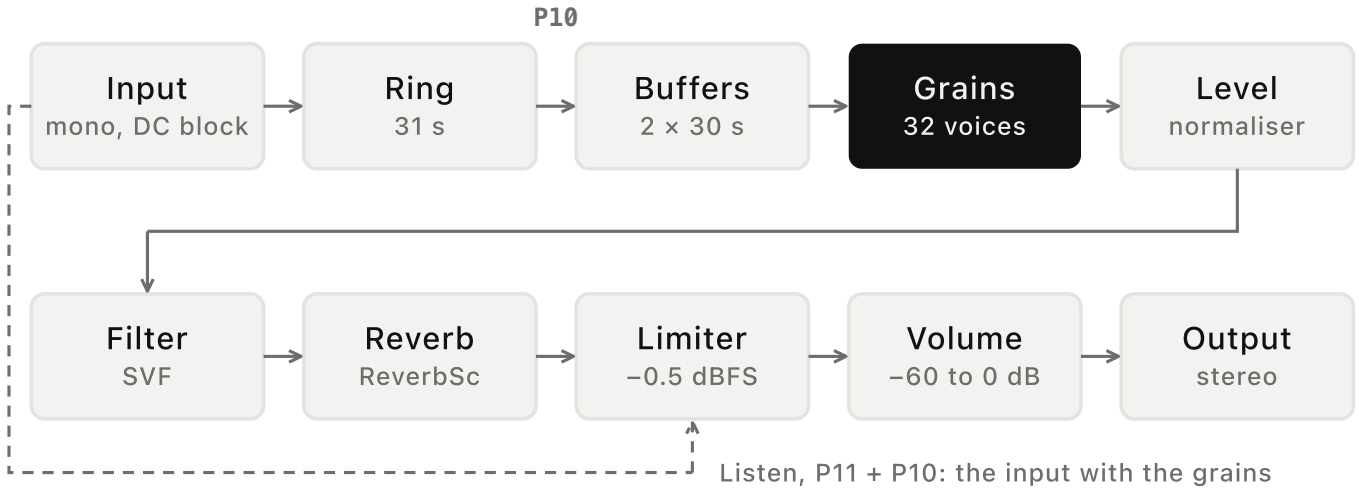
# Engine

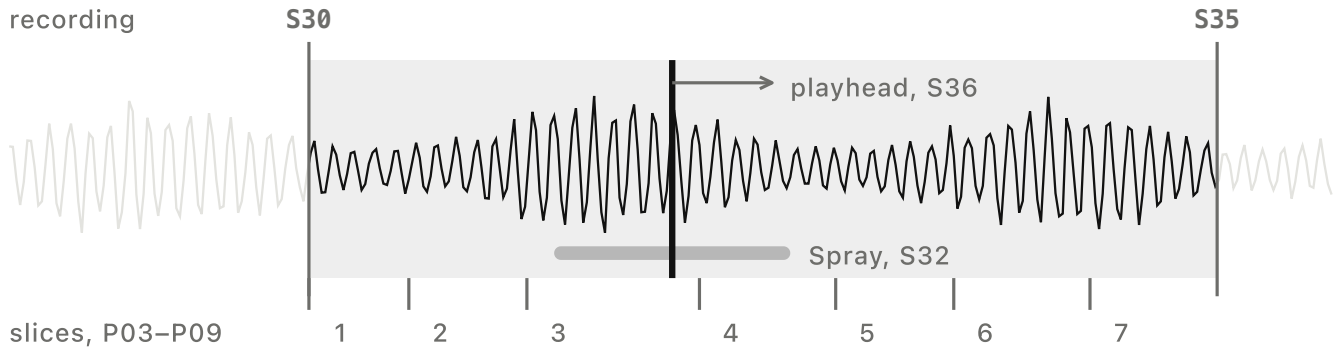
How a recording becomes grains, and how they fall.

## 01 SIGNAL PATH




Both jacks are summed to mono, −3 dB each. The ring takes the input all the time. P10 copies from it into the buffer not playing, and the grains read the buffer playing.

## 02 RECORDING




| | |
|---|---|
| TAP | Under 0.3 s: the 8 s before it. |
| HOLD | From 50 ms before the touch to the release, up to 30 s. |
| TRIM | Silence under −60 dBFS cut from both ends. |
| LAND | Copied at 8× real time. Over 2 s each new grain reads it with rising odds, the rest the recording before. The first rises from silence. |
| SLICES | Its attacks found once, in 5 ms frames, the strongest 64 kept. A loop's seven slices: its start, then its strongest attacks. Too few, as a held note: even parts. |

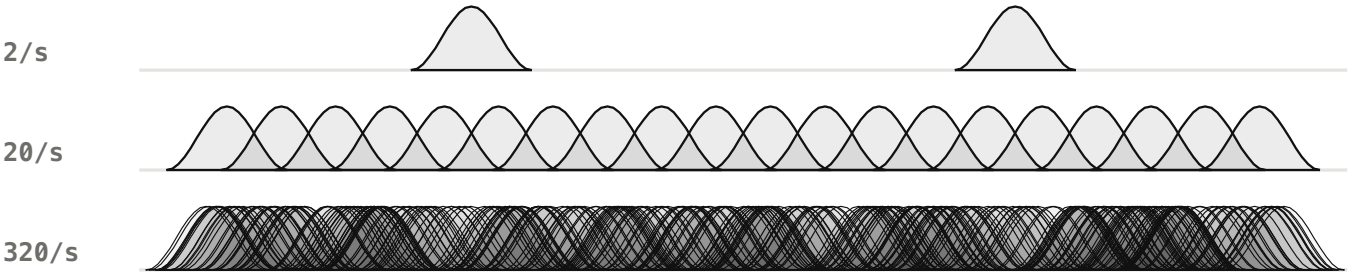
## 03 A GRAIN

| | |
|---|---|
| POSITION | The playhead, and with Spray anywhere round it, evenly, up to the whole loop. |
| START | Moved, at most 5.3 ms or a quarter of its length, to where it continues the grain before in phase: the signs of 512 samples compared, as Clouds' WSOLA player. Not with Spray up, whose grains are meant to differ. |
| READ | Four-point Hermite between samples, as Beads. Rate 2^(n/12): Pitch ±24 semitones, stops at 0, 1, 3, 4, 7, 12, 24. |
| RANDOM PITCH | Up: any pitch to ±12 semitones. Down: octaves, up to two grains in three. |
| SIZE | 5 ms to 2 s, 100 ms at the centre. Random size: part of the way to any size, exponential. |
| WINDOW | Beads' envelope along Shape: square, decay, bell, reverse. At least 0.5 ms up and 10 ms down. |
| PAN | Spray spreads them, wider as more sound at once. Equal power. |
| DIRECTION | The playhead's. Reverse sets the share read the other way. |



square decay bell reverse

## 04 STARTS




Density is grains a second, 0.25 to 320, none at the bottom, whatever their size: grains sounding at once are Density × Size. Random timing takes the starts from a regular pulse to nearly Poisson, the same mean. 32 voices: when all sound, a new grain is dropped.

## 05 LEVEL



in phase
amplitudes add

scattered
powers add

Grains in phase add in amplitude, scattered ones in power. A normaliser measures both every block, 5 ms smoothed, and brings the sum to the input's level with 0.9 headroom, never boosting grains that cancel. Aligned starts keep slowed grains in phase: at ½× a sawtooth's harmonics vary 0.27 dB from one 20 ms frame to the next, as at 1×. Without, 1.0 dB, and 5.6 dB in a dense random cloud.

## 06 TRANSPORT

| | |
|---|---|
| SPEED | 0 to 4×, stops at ¼, ½, 1, 2, 4. The pitch stays. Frozen by S09, the fader scrubs: a full stroke the whole loop. |
| END | At the loop's end, S07: once and stop, turn over and run back, or round again. |
| STOP | P00 held: everything slows to a halt over 0.6 s, as a tape machine, faded over its slowest quarter. Released: back to speed. |
| TRIGGER | P02: the loop from its start (its end in reverse), a grain at once. After Once, it pours again. |
| SLICES | P03–P09 held: each plays its slice with its own playhead, at Speed and Density, the loop waiting. Several at once. |

## 07 EFFECTS

| | |
|---|---|
| FILTER | SVF on the grains. Low-pass 20 kHz to 200 Hz, off at the centre, high-pass 20 Hz to 2 kHz. Resonance 0.7 to 1.5. |
| REVERB | ReverbSc on the grains. Decay 0.5 to 16 s, damped at 10 kHz, up to equal parts with the dry grains. |
| LIMITER | Look-ahead 5 ms, −0.5 dBFS. The grains +6 dB into it. |
| VOLUME | After the limiter: off, then −60 to 0 dB. |

## 08 RUNTIME

| | |
|---|---|
| AUDIO | 48 kHz, 48-sample blocks: 1 ms. Recordings in SDRAM, mono. |
| CPU | 15 % with no grains. 83 % at the most: 32 voices, all slices held, Pitch +24, a recording landing. |
| BROWSER | The same C++ core in WebAssembly, in an AudioWorklet. |